EEG-Based Depression Detection
Using Multiple Machine Learning Techniques

A Complete Beginner-Friendly Explanation + 50 Exam Questions

Based on the research poster by Aditya Krishna (225CS2009), NIT Rourkela — Department of Computer Science and Engineering

PART 1: Complete Explanation (Beginner-Friendly)

1. What Is Depression and Why Is It Hard to Detect?

Depression (also called Major Depressive Disorder or MDD) is a serious mental illness that affects how a person feels, thinks, and handles daily activities. It is NOT just 'feeling sad' — it is a medical condition that changes the brain's chemistry and electrical activity. Traditionally, doctors diagnose depression by talking to patients and using questionnaires (like PHQ-9). But this is subjective — different doctors may give different diagnoses. Scientists wanted a more objective, data-driven method. That is why they turned to EEG (brain signals) and Machine Learning (computers that learn patterns).

2. What Is EEG? (Electroencephalogram)

EEG stands for Electroencephalogram. It is a test that measures the electrical activity in the brain. Just like your heart produces electrical signals (measured by ECG), your brain also produces tiny electrical signals every second. These signals are captured by small sensors called electrodes placed on the scalp (head).

- **Why EEG?**

  The brain of a depressed person shows different electrical patterns compared to a healthy person. EEG can capture these differences non-invasively (without surgery), making it a safe and useful tool.

- **EEG Signal**

  The brain's electrical signal looks like a wavy line on a graph. It changes every millisecond. These waves are grouped into frequency bands (explained below).

- **128-Channel EEG**

  In this study, 128 electrodes were placed on the scalp — each electrode captures signals from a different area of the brain. More electrodes = more detailed brain map.

3. EEG Frequency Bands — What Are They?

The electrical signals from the brain are not all the same speed. Scientists categorize them into bands based on how fast they oscillate (vibrate) per second. This speed is measured in Hz (Hertz = cycles per second).

- **Delta (0.5–4 Hz)**

  Very slow waves. Dominant during deep, dreamless sleep. In depression, delta activity may be abnormal during waking hours.

- **Theta (4–8 Hz)**

  Slow waves. Associated with drowsiness, light sleep, and emotional processing. Often elevated in depressed patients.

- **Alpha (8–13 Hz)**

  Medium waves. Seen when you are awake but relaxed with eyes closed. Depressed patients often show asymmetry (left vs. right brain alpha difference).

- **Beta (13–30 Hz)**

  Fast waves. Associated with active thinking, alertness. Abnormal beta patterns are seen in anxiety and depression.

- **Why does this matter?**

  Different bands tell us different things about brain activity. The study used all these bands together as features (inputs) for the machine learning models.

4. The MODMA Dataset — Where Did the Data Come From?

You cannot build a machine learning model without data. This study used the MODMA dataset — a publicly available collection of EEG recordings from real patients.

- **MODMA**

  Multi-Modal Open Dataset for Mental-disorder Analysis. It contains EEG data recorded from real human participants.

- **Participants**

  53 total: 24 people diagnosed with MDD (patients) and 29 healthy controls (normal people). Eyes were kept closed during recording to reduce visual artefacts.

- **Equipment**

  Two types of recorders were used: (1) A 128-electrode elastic cap for detailed brain mapping, and (2) A wearable 3-electrode collector for portable/practical use.

- **Inclusion Criteria**

  Who could participate: Age 18–55 years; PHQ-9 score ≥ 5 (indicating depression); no psychotropic (mind-altering) drugs in the last 2 weeks; gave written consent.

- **Exclusion Criteria**

  Who was excluded: People with brain damage, history of other mental disorders, alcohol/drug dependence, or pregnancy.

- **PHQ-9**

Patient Health Questionnaire-9. A standard 9-question survey used to screen for depression. Higher score = more severe depression. Score ≥ 5 suggests at least mild depression.

- **ANOVA Test**

  Analysis of Variance — a statistical test. It confirmed that the age difference between MDD and healthy groups was NOT significant ($p > 0.05$), meaning the EEG differences we see are due to depression, not age.

5. MDD Subtypes — The 6 Categories of Depression

Depression is not a single disorder — it comes in different forms. The MODMA dataset classified MDD patients into 6 subtypes:

- **Obsessive-Compulsive (OCD)**

  Unwanted repetitive thoughts and compulsive behaviors (e.g., washing hands repeatedly). Treated as a depressive spectrum disorder here.

- **Addictive Disorder**

  Depression linked to or caused by substance/behavioral addiction (alcohol, drugs, gambling).

- **Trauma and Stress-Related**

  Depression triggered by traumatic events — accidents, abuse, PTSD, grief.

- **Mood Disorder**

  Core mood disturbances — persistent sadness, hopelessness, loss of interest in life.

- **Schizophrenia**

  A severe mental disorder with hallucinations and delusions, often co-occurring with depression.

- **Anxiety Disorder**

  Chronic excessive worry and fear co-existing with depression.

- **Healthy Control**

  People with no mental health diagnosis — used as the comparison group.

6. Pre-Processing Pipeline — Cleaning the Brain Data

Raw EEG signals are very noisy. Before any analysis, the data must be cleaned. Think of it like washing vegetables before cooking. The pipeline has 5 steps:

- **Step 1 — ICA (Independent Component Analysis) Artifact Removal**

  ICA is a mathematical technique that separates different 'sources' mixed in the EEG signal. Artefacts (noise) come from: eye blinks, muscle movements, heartbeat. ICA identifies and removes these non-brain signals, keeping only pure brain activity.

- **Step 2 — Re-Referencing**

  EEG signals are measured relative to a reference point. Re-referencing means subtracting a common average from all channels to reduce shared noise and improve the signal-to-noise ratio (SNR). Higher SNR = cleaner signal.

- **Step 3 — Band-pass Filter (50 Hz)**

A filter that allows only signals between certain frequencies to pass through. Here, signals above 50 Hz are removed because 50 Hz is the AC power-line frequency (electrical interference from power outlets). The filtered signal is then converted into a NumPy array (a Python data structure for numerical computation).

- **Step 4 — Feature Extraction**

  Features are the measurable properties extracted from the EEG signal that the ML model will learn from. Two types: (a) Linear features — e.g., mean, variance, power in each band. (b) Non-linear features — e.g., Spectral Entropy (how disordered/unpredictable the signal is), SVD Entropy (Singular Value Decomposition Entropy, measures signal complexity). Demographic data (age, gender) is also merged with EEG features.

- **Step 5 — Signal Segmentation**

  The continuous EEG recording (which can be minutes long) is divided into fixed-length epochs (small time windows, e.g., 1-second or 2-second chunks). This gives many samples from one recording, making the dataset larger and more useful for training.

7. Machine Learning Models — The Three Classifiers

After extracting features, three ML models were trained to classify EEG data into MDD subtypes vs. healthy control. Each model learned from the features and made predictions.

- **Random Forest (RF)**

  Imagine asking 100 different doctors for an opinion, then going with the majority vote. That's Random Forest. It builds many Decision Trees (each trained on a random subset of data — called Bootstrap Sampling). Each tree votes, and the majority class wins. Estimators tested: 100, 300, 500 trees. More trees generally = more accurate but slower.

- **Decision Tree**

  A flowchart-like model that splits data based on feature values (e.g., 'Is Alpha power > 0.5? If yes → MDD, else → Healthy').

- **XGBoost (Extreme Gradient Boosting)**

  A powerful algorithm where trees are built sequentially — each new tree tries to fix the errors made by the previous one. This is called Gradient Boosting. 'Extreme' because it adds regularization (penalty for complexity) to prevent overfitting. Sub-sample rates tested: 0.3, 0.5, 1.0 (fraction of data used per tree). Very popular in competitions due to high accuracy.

- **1D-CNN (1-Dimensional Convolutional Neural Network)**

  A deep learning model inspired by how the brain processes information in layers. Designed for sequential/time-series data like EEG. Architecture: Conv1D (kernel=11, ReLU) → captures local patterns in the signal over 11 time points, ReLU activation = keeps positive values, zeroes negatives. Dropout (0.2) → randomly deactivates 20% of neurons during training to prevent overfitting. MaxPooling (size=4) → compresses the signal by keeping the maximum value in each 4-point window, reducing size while keeping important features. Dense (100 units, ReLU) → a fully connected layer that combines learned features. Sigmoid Output → gives a probability between 0 and 1 (e.g., 0.92 = 92% likely to be MDD).

- **Overfitting**

  When a model memorizes training data too well and fails on new data. Like a student who memorizes answers but cannot solve new problems. Dropout and regularization prevent this.

- **10-Fold Cross-Validation**

  The data is split into 10 equal parts. The model trains on 9 parts and tests on 1 part, repeating 10 times so every part gets tested. This gives a reliable estimate of model performance on unseen data and reduces bias.

8. CNN Training Details

The CNN was trained with these settings:

- **Epochs = 25**

  One epoch = one full pass through all training data. The CNN was trained for 25 epochs (25 full passes). Accuracy improved from 87.9% at epoch 1 to 99.3% at epoch 25.

- **Batch Size = 32**

  Instead of feeding all data at once, data is fed in batches of 32 samples. This makes training faster and more stable.

- **Learning Rate = 0.001**

  Controls how much the model adjusts its weights after each batch. Too high = unstable learning; too low = very slow learning. 0.001 is a standard starting point.

- **Binary Cross-Entropy Loss**

  The loss function measures how wrong the model's predictions are. Binary Cross-Entropy is used when there are 2 classes (MDD vs. Healthy). The model tries to minimize this loss during training.

- **Train vs. Validation Curves**

  The accuracy and loss curves for training data and validation data (unseen during training) converge — meaning no overfitting occurred.

9. Evaluation Metrics — How Was Performance Measured?

Just saying '97% accuracy' is not enough. These metrics give a fuller picture:

- **Accuracy**

  Percentage of total correct predictions. (Correct predictions / Total predictions) × 100.

- **Precision**

  Of all samples predicted as MDD, what fraction was actually MDD? High precision = fewer false alarms.

- **Recall (Sensitivity)**

  Of all actual MDD cases, what fraction did the model correctly find? High recall = fewer missed cases. In medical diagnosis, recall is very important — missing a sick person is dangerous.

- **F1 Score**

  Harmonic mean of Precision and Recall. Balances both. Reported as Micro (overall) and Macro (average per class).

- **ROC-AUC**

  Receiver Operating Characteristic — Area Under Curve. A value of 1.0 = perfect classifier; 0.5 = random guessing. Measures how well the model distinguishes between classes at various thresholds.

10. Results — How Well Did Each Model Perform?

CNN outperformed both RF and XGBoost on ALL 6 MDD subtypes. Here are the accuracy scores:

- **Addictive Disorder**
  RF: 0.82, XGBoost: 0.85, CNN: 0.94
- **Trauma/Stress**
  RF: 0.73, XGBoost: 0.82, CNN: 0.91
- **Mood Disorder**
  RF: 0.76, XGBoost: 0.81, CNN: 0.93
- **Schizophrenia**
  RF: 0.72, XGBoost: 0.80, CNN: 0.93
- **Anxiety Disorder**
  RF: 0.73, XGBoost: 0.77, CNN: 0.94
- **OCD**
  RF: 0.63, XGBoost: 0.69, CNN: 0.96 — CNN's biggest improvement here!
- **Overall Best**
  CNN achieved up to 97% accuracy at 25 epochs. CNN won because it automatically learns spatial and temporal patterns in the EEG signal without manual feature engineering.

11. Conclusion and Future Work

The study proved that EEG + Deep Learning (CNN) can detect depression with very high accuracy. Key takeaways:

- **Why CNN won**
  CNNs learn hierarchical features automatically from raw signals, making them superior to traditional ML on time-series data.
- **Demographic integration**
  Adding age and gender data alongside EEG features improved accuracy.
- **Future: More Data**
  The dataset only had 53 participants. More data = better generalization.
- **Future: Severity Grading**
  Instead of just 'MDD vs. Healthy', grade severity: mild, moderate, severe.
- **Future: Ear-BCI Electrodes**
  Replace the 128-electrode lab cap with lightweight ear-worn electrodes for everyday use.
- **Future: Explainable AI (XAI)**
  Make the model explain its decisions in human terms so doctors can trust it.
- **Future: Real-Time Monitoring**

Deploy the model on edge devices (smartphones, smartwatches) for continuous mental health monitoring.

PART 2: 50 Possible Exam Questions with Answers

These questions are organized by topic and cover all aspects of the poster — from basic concepts to in-depth technical understanding.

SECTION A: Basic Concepts

Q1. What does EEG stand for and what does it measure?

Answer: EEG stands for Electroencephalogram. It measures the electrical activity (brain waves) generated by neurons in the brain, captured via electrodes placed on the scalp. It is non-invasive and records signals in real-time.

Q2. What is Major Depressive Disorder (MDD)?

Answer: MDD is a serious mental health condition characterized by persistent sadness, loss of interest, fatigue, and cognitive impairment. Unlike normal sadness, MDD is a clinical diagnosis that alters brain chemistry and electrical patterns, requiring medical treatment.

Q3. Why is traditional diagnosis of depression considered subjective?

Answer: Traditional diagnosis relies on clinical interviews and self-report questionnaires (like PHQ-9), which depend on the patient's honest responses and the clinician's interpretation. Different doctors may reach different conclusions, making it prone to variability and bias.

Q4. Why is EEG used to detect depression instead of other brain imaging methods?

Answer: EEG is non-invasive, relatively low-cost, portable, and has high temporal resolution (captures brain activity millisecond by millisecond). Unlike fMRI (which is expensive and requires a large machine), EEG can capture dynamic electrical changes associated with depression.

Q5. What are EEG frequency bands? Name all four studied here.

Answer: EEG signals are categorized by oscillation speed (Hz): Delta (0.5–4 Hz) — deep sleep; Theta (4–8 Hz) — drowsiness/emotions; Alpha (8–13 Hz) — relaxed wakefulness; Beta (13–30 Hz) — active thinking/alertness.

SECTION B: MODMA Dataset

Q6. What does MODMA stand for and what does it contain?

Answer: MODMA stands for Multi-Modal Open Dataset for Mental-disorder Analysis. It contains resting-state EEG recordings from 24 MDD patients and 29 healthy controls (53 total), recorded with a 128-electrode cap and a wearable 3-electrode device.

Q7. How many MDD patients and healthy controls were in the dataset?

Answer: 24 MDD patients and 29 healthy controls, totaling 53 participants. Both groups had similar age distributions, confirmed by ANOVA ($p > 0.05$).

Q8. What is the PHQ-9 and how was it used in participant selection?

Answer: PHQ-9 (Patient Health Questionnaire-9) is a standardized 9-item screening tool for depression. A score of ≥5 was required for inclusion as an MDD participant in this study, ensuring clinical confirmation of depression.

Q9. Why were participants asked to keep their eyes closed during EEG recording?

Answer: To reduce visual artefacts — electrical noise caused by eye movements and blinking. Eyes-closed resting-state recording produces cleaner, more consistent brain signals.

Q10. What were the inclusion criteria for the MODMA dataset?

Answer: Age 18–55 years; PHQ-9 score ≥ 5; no psychotropic medications in the last 2 weeks; provided written informed consent.

Q11. What were the exclusion criteria for the dataset?

Answer: Brain damage history; prior mental disorder diagnosis (other than MDD); alcohol or drug dependence; pregnancy.

Q12. What was the purpose of the ANOVA test on participant age?

Answer: ANOVA (Analysis of Variance) confirmed that there was no statistically significant age difference between MDD and healthy groups ($p > 0.05$). This means observed EEG differences reflect depression, not age-related brain changes.

Q13. What two types of EEG equipment were used in MODMA?

Answer: (1) A 128-electrode elastic cap — for detailed, high-density brain mapping in a lab setting. (2) A wearable 3-electrode collector — for portable, real-world use. The study's ML pipeline was designed to work with both.

SECTION C: MDD Subtypes

Q14. List the 6 MDD subtypes used in this study.

Answer: 1. Obsessive-Compulsive Disorder (OCD), 2. Addictive Disorder, 3. Trauma and Stress-Related Disorder, 4. Mood Disorder, 5. Schizophrenia, 6. Anxiety Disorder.

Q15. Why is classifying MDD into subtypes important?

Answer: Different subtypes of depression have different brain activity patterns. Subtype-specific classification allows doctors to provide more targeted, precise treatment rather than a one-size-fits-all approach, improving patient outcomes.

Q16. Which MDD subtype showed the highest CNN accuracy and which showed the lowest for traditional models?

Answer: OCD showed the highest CNN accuracy (0.96) and the lowest RF accuracy (0.63), demonstrating CNN's greatest improvement margin. Addictive Disorder had the highest RF and XGBoost scores among subtypes.

SECTION D: Pre-Processing Pipeline

Q17. What is ICA and why is it used in EEG pre-processing?

Answer: ICA (Independent Component Analysis) is a statistical technique that separates mixed signals into independent sources. In EEG, it separates brain-generated signals from artefacts caused by eye blinks, muscle movements, and heartbeat, giving cleaner data.

Q18. What is Re-Referencing and why is it important?

Answer: Re-referencing recalculates EEG signals relative to a common average of all electrodes, minimizing noise that is not related to brain activity. It improves the signal-to-noise ratio (SNR), making features more reliable.

Q19. Why is a 50 Hz band-pass filter applied to the EEG signal?

Answer: 50 Hz is the AC power-line frequency in many countries. Without filtering, this electrical interference from power sources contaminates the EEG signal. The filter removes frequencies above 50 Hz, retaining only brain-relevant signals.

Q20. What are linear and non-linear features in EEG analysis?

Answer: Linear features: statistical measures like mean, variance, and frequency band power — easy to compute. Non-linear features: complexity measures like Spectral Entropy (signal unpredictability) and SVD Entropy (signal information content) — capture brain dynamics that linear features miss.

Q21. What is Spectral Entropy and what does it indicate?

Answer: Spectral Entropy measures how uniform or disordered the power distribution across EEG frequency bands is. High entropy = more disordered/complex signal. Depressed brains often show abnormal entropy values compared to healthy brains.

Q22. What is Signal Segmentation and why is it done?

Answer: Segmentation divides a long continuous EEG recording into smaller fixed-length time windows (epochs). This creates many samples from one recording, increases dataset size, and allows consistent windowed feature extraction across all 128 channels.

Q23. What is a NumPy array and why is the EEG signal converted to it?

Answer: NumPy (Numerical Python) arrays are efficient multi-dimensional data structures used in Python for mathematical computation. Converting EEG signals to NumPy arrays makes them compatible with Python ML libraries (scikit-learn, TensorFlow, Keras).

SECTION E: Machine Learning Models

Q24. Explain how Random Forest works in simple terms.

Answer: Random Forest builds many decision trees, each trained on a random sample of data (bootstrap sampling). Each tree independently classifies the input, and the final prediction is the class with the most votes (majority voting). More trees reduce variance and improve generalization.

Q25. What is Bootstrap Sampling?

Answer: Bootstrap sampling creates multiple training datasets by randomly sampling (with replacement) from the original dataset. This means the same data point can appear multiple times in one sample, and some points may be absent. It ensures each tree in the forest sees a different data perspective.

Q26. How does XGBoost differ from Random Forest?

Answer: Random Forest builds trees in parallel (independently), while XGBoost builds trees sequentially — each new tree corrects the errors of the previous one (gradient boosting). XGBoost also adds regularization to prevent overfitting, making it more accurate on complex data.

Q27. What does 'gradient boosting' mean in XGBoost?

Answer: Gradient boosting means the algorithm trains new trees to minimize the residual error (mistakes) of the current ensemble. It uses gradient descent — a mathematical optimization method — to minimize the loss function, sequentially improving predictions.

Q28. What is the architecture of the 1D-CNN used in this study?

Answer: Input → Conv1D (kernel=11, ReLU activation) → Dropout(0.2) → MaxPooling(size=4) → Dense(100 units, ReLU) → Sigmoid output. This processes the EEG time-series through convolution, regularization, downsampling, and classification layers.

Q29. What is the role of Conv1D in a 1D-CNN?

Answer: Conv1D (1-Dimensional Convolution) applies a sliding window (kernel) of size 11 across the EEG time series to detect local patterns — like repeated wave shapes associated with depression. ReLU activation introduces non-linearity, enabling the model to learn complex patterns.

Q30. What is Dropout and why is it set to 0.2?

Answer: Dropout randomly deactivates 20% of neurons during each training step. This prevents neurons from co-adapting (relying on each other), forcing the network to learn more robust, generalizable features. Rate 0.2 (20%) is a common starting value that balances regularization and learning capacity.

Q31. What does MaxPooling do in the CNN?

Answer: MaxPooling reduces the spatial/temporal size of the feature map by taking the maximum value in each non-overlapping window of size 4. This reduces dimensionality, computation, and retains the most prominent (dominant) features while discarding minor fluctuations.

Q32. What is the Dense layer and what does it do?

Answer: The Dense layer (fully connected layer) has 100 neurons, where every neuron is connected to all inputs from the previous layer. It combines the abstract features learned by Conv1D and MaxPooling into higher-level representations used for final classification.

Q33. Why is Sigmoid used as the output activation instead of Softmax?

Answer: Sigmoid outputs a probability between 0 and 1 for binary classification (MDD vs. Healthy). Softmax is used for multi-class classification. Since each subtype is classified as a binary problem (MDD subtype vs. healthy), Sigmoid is appropriate.

Q34. What is Binary Cross-Entropy loss?

Answer: Binary Cross-Entropy measures the difference between the true binary label (0 or 1) and the predicted probability. It penalizes confident wrong predictions heavily. The CNN minimizes this loss during training using backpropagation.

Q35. What is 10-fold Cross-Validation and why is it important?

Answer: The dataset is split into 10 equal folds. The model trains on 9 folds and tests on 1 fold, repeated 10 times with a different test fold each time. This provides 10 performance estimates whose average is more reliable than a single train/test split. It also reduces overfitting bias.

Q36. What is overfitting and how was it prevented in this study?

Answer: Overfitting is when a model learns training data too specifically and performs poorly on new data. It was prevented by: Dropout (0.2) in the CNN, Regularization in XGBoost, Bootstrap Sampling in Random Forest, and 10-fold Cross-Validation.

SECTION F: Training and Results

Q37. What was the CNN's accuracy at epoch 1 and epoch 25?

Answer: Epoch 1: 87.9% accuracy, 0.323 loss. Epoch 25: 99.3% accuracy, 0.043 loss. The steady improvement without divergence between training and validation curves confirmed no overfitting.

Q38. What does it mean when training and validation curves converge?

Answer: Convergence means the model performs similarly on both training data (data it learned from) and validation data (unseen data). This confirms the model has learned generalizable patterns, not just memorized training examples.

Q39. Which model performed best overall and by how much?

Answer: CNN outperformed both RF and XGBoost on all 6 MDD subtypes. For example, on OCD: RF=0.63, XGBoost=0.69, CNN=0.96 — a 33% improvement over RF. Overall CNN achieved up to 97% accuracy.

Q40. What hyperparameters were tuned for Random Forest?

Answer: The number of estimators (trees): 100, 300, and 500 were tested. The best configuration was selected based on cross-validation accuracy.

Q41. What hyperparameters were tuned for XGBoost?

Answer: Sub-sample rate: 0.3, 0.5, and 1.0 were tested. Sub-sample controls the fraction of training data used for each tree — lower values add randomness to reduce overfitting.

Q42. What is ROC-AUC and what would a perfect value be?

Answer: ROC-AUC (Receiver Operating Characteristic — Area Under Curve) measures a classifier's ability to distinguish between classes across all decision thresholds. AUC = 1.0 is perfect; AUC = 0.5 is equivalent to random guessing.

SECTION G: Features and Methodology

Q43. How were demographic features integrated with EEG features?

Answer: After extracting linear and non-linear EEG features, age and gender values were appended to the feature vector before feeding into the ML models. This gave the models additional context that could correlate with depression patterns.

Q44. What is SVD Entropy?

Answer: SVD Entropy (Singular Value Decomposition Entropy) measures the complexity/information content of the EEG signal matrix. It uses the singular values from SVD decomposition to compute an entropy measure, reflecting signal structure and predictability.

Q45. Why did the CNN outperform RF and XGBoost on EEG data?

Answer: CNNs can automatically learn spatial and temporal feature hierarchies from raw or minimally processed signals without manual feature engineering. EEG data has rich temporal patterns that CNNs capture through convolution. RF and XGBoost depend on hand-crafted features, limiting their expressiveness.

SECTION H: Future Work and Applications

Q46. What is Explainable AI (XAI) and why is it needed in this context?

Answer: XAI refers to AI methods that can explain their predictions in human-understandable terms. In medical applications, doctors must understand why a model made a diagnosis to trust and use it clinically. Without explainability, 'black box' models may be rejected by clinicians despite high accuracy.

Q47. What are Ear-BCI electrodes and why are they proposed for future work?

Answer: Ear-BCI (Brain-Computer Interface) electrodes are small, wearable sensors placed in/around the ear to capture EEG signals without a full 128-channel cap. They make EEG recording practical for daily life, enabling real-world depression monitoring outside clinical settings.

Q48. What is Edge Computing in the context of real-time mental health monitoring?

Answer: Edge Computing processes data locally on the device (smartphone, smartwatch) rather than sending it to a remote server. This enables real-time, low-latency EEG analysis with privacy preserved, making continuous mental health monitoring feasible on wearable devices.

Q49. Why does the study propose severity grading as future work?

Answer: The current model only detects MDD vs. Healthy (binary). Clinical treatment depends on severity — mild, moderate, or severe depression require different interventions. Adding severity grading would make the system clinically actionable and more useful for personalized treatment planning.

Q50. What are the main limitations of this study and how could they be addressed?

Answer: Main limitations: (1) Small dataset (53 participants) — addressed by collecting more data from diverse populations. (2) Lab-based 128-channel recording — addressed by developing wearable ear-BCI solutions. (3) Black-box model — addressed by Explainable AI techniques. (4) Binary classification only — addressed by adding severity grading. (5) Static analysis — addressed by real-time edge computing deployment.

Source: Ksibi et al., Diagnostics 2023, 13, 1779. doi:10.3390/diagnostics13101779 | Poster by Aditya Krishna (225CS2009), NIT Rourkela